Multivariate Data Analysis

CH5440

Assignment 2

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CH24B108

Source code: https://github.com/orcus108/ch5440-mvda-assignment2

Contents

1 Question 1: Quadratic Polynomial Regression **3**
1.1 (i) Raw Design Matrix . . . 3
1.2 (ii) Step-Size Coded Matrix . . . 3
1.3 (iii) Min-Max Scaled Matrix . . . 4
1.4 Case 1: Raw Predictors (X) . . . 4
1.4.1 (iv) Regression Coefficients . . . 4
1.4.2 (v) Sum of Squares . . . 4
1.4.3 (vi) Variance-Covariance Matrix . . . 4
1.4.4 (vii) Model Fit Statistics . . . 5
1.5 Case 2: Step-Size Coded Predictors (x) . . . 5
1.5.1 (iv) Regression Coefficients . . . 5
1.5.2 (v) Sum of Squares . . . 5
1.5.3 (vi) Variance-Covariance Matrix . . . 6
1.5.4 (vii) Model Fit Statistics . . . 6
1.6 Case 3: Min-Max Scaled Predictors (z) . . . 6
1.6.1 (iv) Regression Coefficients . . . 6
1.6.2 (v) Sum of Squares . . . 6
1.6.3 (vi) Variance-Covariance Matrix . . . 6
1.6.4 (vii) Model Fit Statistics . . . 6
1.7 Summary Comparison Across Transformations . . . 7

2 Question 2: Mixture Regression: Yarn Elongation **8**
2.1 (a) Mixture vs. Conventional Regression . . . 8
2.2 (b) Design Matrix and Response Vector . . . 8
2.3 (c) Estimated Parameters . . . 9
2.4 (d & f) Sum of Squares (Matrix vs. Summation) . . . 9
2.5 (e) Source of Error and Pure Error . . . 9
2.6 (g) Variance-Covariance Matrix . . . 9
2.7 (h) Model Fit Statistics and Predicted vs. Observed . . . 10
2.8 Constrained Optimisation Method . . . 10
2.9 (i) Optimal Composition: Maximum Elongation . . . 11
2.10 (j) Optimal Composition: Minimum Elongation . . . 11
2.11 (k) Validation . . . 11

3 Question 3: Logistic Regression: Experience vs. Hiring **14**
3.1 MLE Methodology: Newton-Raphson . . . 14
3.2 Estimated Parameters . . . 14
3.3 Predicted vs. Observed . . . 15
3.4 Classification at Threshold 0.5 . . . 15
3.5 Logistic Regression Curve . . . 16
3.6 JMP Validation . . . 16

4 Question 4: Logistic Regression: Income and Gender (Purchase Decision) **19**
4.1 (a) Model 1: Income Only . . . 19
4.2 (b) Model 2: Income + Gender . . . 20
4.3 Standard Errors and Odds Ratios . . . 21

4.4 Classification at Threshold 0.5 22
4.5 Likelihood Ratio Test: Model 2 vs. Model 1 22
4.6 Newton-Raphson Convergence 22
4.7 (c) JMP Validation 23

5 Question 5: Grouped Binomial MLE: Coupon Redemption 25
5.1 Newton-Raphson Iterations 25
5.2 Estimated Parameters 25
5.3 Predicted vs. Observed Proportions 26
5.4 Goodness-of-Fit 26
5.5 Plots 26

6 Question 6: Bernoulli MLE vs. Grouped Binomial 28
6.1 Newton-Raphson Iterations: Bernoulli MLE 28
6.2 Estimated Parameters 28
6.3 Predicted vs. Observed (by Income Group) 28
6.4 Comparison: Bernoulli vs. Grouped Binomial 29
6.5 Newton-Raphson Convergence 29

7 Question 7: Multiple Logistic Regression: Dengue Disease (Kutner CH14) 30
7.1 Data Summary 30
7.2 Newton-Raphson Iterations 30
7.3 Estimated Model Parameters 30
7.4 Predicted Probabilities (First 20 Observations) 31
7.5 Model Significance (Likelihood Ratio Test vs. Null) 31
7.6 Interpretation 31
7.7 Newton-Raphson Convergence 32

1 Question 1: Quadratic Polynomial Regression

Data: $n = 11$ observations, $\bar{Y} = 171.7273$.

The full second-order polynomial model in two predictors X_1 and X_2 is:

$$Y = b_0 + b_1 X_1 + b_2 X_2 + b_{11} X_1^2 + b_{22} X_2^2 + b_{12} X_1 X_2 + \varepsilon$$

Design matrix columns: $\left[1,\ X_1,\ X_2,\ X_1^2,\ X_2^2,\ X_1 X_2\right]$.

Parameters are estimated by ordinary least squares (OLS) via the normal equations:

$$\hat{\beta} = (X'X)^{-1} X'y$$

Fitted values are $\hat{y} = X\hat{\beta}$ and residuals are $e = y - \hat{y}$. The sums of squares follow as:

$$\text{SST} = y'y - n\bar{y}^2, \qquad \text{SSR} = \hat{\beta}' X' y - n\bar{y}^2, \qquad \text{SSE} = (y - \hat{y})'(y - \hat{y}) = e'e$$

1.1 (i) Raw Design Matrix

No transformation is applied; raw predictor values are used directly. Representative rows of the 11×6 matrix are shown below (full matrix in `out1.txt`).

Table 1: Raw design matrix X_{raw}: representative rows

1	X_1	X_2	X_1^2	X_2^2	$X_1 X_2$
1	0.6	10	0.36	100	6
1	1.0	10	1.00	100	10
1	1.4	10	1.96	100	14
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1	1.0	30	1.00	900	30
1	1.4	30	1.96	900	42

1.2 (ii) Step-Size Coded Matrix

Centring and scaling by the step size (half the experimental range):

$$x_j = \frac{X_j - \mu_j}{\text{step}_j}, \qquad \mu_{X_1} = 1.0,\ \text{step}_{X_1} = 0.4, \qquad \mu_{X_2} = 20.0,\ \text{step}_{X_2} = 10.0$$

Table 2: Step-size coded matrix X_{cod}: representative rows

1	x_1	x_2	x_1^2	x_2^2	$x_1 x_2$
1	-1	-1	1	1	1
1	0	-1	0	1	0
1	1	-1	1	1	-1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1	0	1	0	1	0
1	1	1	1	1	1

1.3 (iii) Min-Max Scaled Matrix

Scaling by the full experimental range:

$$z_j = \frac{X_j - \mu_j}{\max_j - \min_j}, \qquad \text{range}_{X_1} = 0.8, \qquad \text{range}_{X_2} = 20.0$$

Table 3: Min-max scaled matrix X_{mm}: representative rows

1	z_1	z_2	z_1^2	z_2^2	$z_1 z_2$
1	-0.5	-0.5	0.25	0.25	0.25
1	0	-0.5	0	0.25	0
1	0.5	-0.5	0.25	0.25	-0.25
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1	0	0.5	0	0.25	0
1	0.5	0.5	0.25	0.25	0.25

1.4 Case 1: Raw Predictors (X)

1.4.1 (iv) Regression Coefficients

Table 4: Estimated coefficients: raw predictors

Parameter	Estimate
b_0 (intercept)	279.822368
b_1 (X_1)	-452.763158
b_2 (X_2)	10.031579
b_{11} (X_1^2)	143.256579
b_{22} (X_2^2)	-0.135789
b_{12} ($X_1 X_2$)	2.500000

1.4.2 (v) Sum of Squares

Table 5: ANOVA sum of squares: raw (matrix and summation approaches agree)

Source	Matrix	Summation
SST	49238.1818	49238.1818
SSR	45102.4976	45102.4976
SSE	4135.6842	4135.6842

1.4.3 (vi) Variance-Covariance Matrix

$s^2 = 827.1368$. The full 6×6 variance-covariance matrix $s^2(A'A)^{-1}$ (entries rounded to 2 d.p.; machine-zero entries shown as 0):

$$\begin{bmatrix} 17747.62 & -25553.27 & -538.55 & 9761.03 & 6.31 & 258.48 \\ -25553.27 & 57047.05 & -176.85 & -25507.92 & 10.88 & -258.48 \\ -538.55 & -176.85 & 66.54 & 217.67 & -1.31 & -12.92 \\ 9761.03 & -25507.92 & 217.67 & 12753.96 & -5.44 & 0 \\ 6.31 & 10.88 & -1.31 & -5.44 & 0.03 & 0 \\ 258.48 & -258.48 & -12.92 & 0 & 0 & 12.92 \end{bmatrix}$$

Condition number of $(A'A)$: $\kappa = 3.1842 \times 10^8$.

1.4.4 (vii) Model Fit Statistics

Table 6: Model fit statistics: raw

Statistic	Value
R^2	0.916007
Adj. R^2	0.832013
PRESS	35532.4507
R^2_{PRESS}	0.278356

PRESS (Predicted Residual Error Sum of Squares) is computed via the hat-matrix diagonal $h_{ii} = [H]_{ii}$, where $H = X(X'X)^{-1}X'$, without refitting the model:

$$\text{PRESS} = \sum_{i=1}^{n} \left(\frac{e_i}{1 - h_{ii}} \right)^2$$

Each term is the leave-one-out prediction error for observation i. $R^2_{\text{PRESS}} = 1 - \text{PRESS}/\text{SST}$ measures out-of-sample predictive power; the large gap between $R^2 = 0.916$ and $R^2_{\text{PRESS}} = 0.278$ signals overfitting for this small dataset ($n = 11$, $p = 6$).

1.5 Case 2: Step-Size Coded Predictors (x)

1.5.1 (iv) Regression Coefficients

Table 7: Estimated coefficients: step-size coded

Parameter	Estimate
b_0 (intercept)	166.631579
b_1 (x_1)	−46.500000
b_2 (x_2)	71.000000
b_{11} (x_1^2)	22.921053
b_{22} (x_2^2)	−13.578947
b_{12} (x_1x_2)	10.000000

1.5.2 (v) Sum of Squares

Identical to Case 1: SST = 49238.1818, SSR = 45102.4976, SSE = 4135.6842. Matrix and summation approaches agree.

1.5.3 (vi) Variance-Covariance Matrix

$s^2 = 827.1368$. The 6×6 variance-covariance matrix:

$$\begin{bmatrix} 217.6676 & 0 & 0 & -130.6006 & -130.6006 & 0 \\ 0 & 137.8561 & 0 & 0 & 0 & 0 \\ 0 & 0 & 137.8561 & 0 & 0 & 0 \\ -130.6006 & 0 & 0 & 326.5014 & -87.0670 & 0 \\ -130.6006 & 0 & 0 & -87.0670 & 326.5014 & 0 \\ 0 & 0 & 0 & 0 & 0 & 206.7842 \end{bmatrix}$$

Condition number of $(A'A)$: $\kappa = 9.5000$.

1.5.4 (vii) Model Fit Statistics

All statistics identical to Case 1: $R^2 = 0.9160$, Adj. $R^2 = 0.8320$, PRESS $= 35532.4507$, $R^2_{\text{PRESS}} = 0.2784$.

1.6 Case 3: Min-Max Scaled Predictors (z)

1.6.1 (iv) Regression Coefficients

Table 8: Estimated coefficients: min-max scaled

Parameter	Estimate
b_0 (intercept)	166.631579
b_1 (z_1)	−93.000000
b_2 (z_2)	142.000000
b_{11} (z_1^2)	91.684211
b_{22} (z_2^2)	−54.315789
b_{12} ($z_1 z_2$)	40.000000

1.6.2 (v) Sum of Squares

Identical to Cases 1 and 2: SST $= 49238.1818$, SSR $= 45102.4976$, SSE $= 4135.6842$.

1.6.3 (vi) Variance-Covariance Matrix

$s^2 = 827.1368$. The 6×6 variance-covariance matrix:

$$\begin{bmatrix} 217.6676 & 0 & 0 & -522.4022 & -522.4022 & 0 \\ 0 & 551.4246 & 0 & 0 & 0 & 0 \\ 0 & 0 & 551.4246 & 0 & 0 & 0 \\ -522.4022 & 0 & 0 & 5224.0222 & -1393.0726 & 0 \\ -522.4022 & 0 & 0 & -1393.0726 & 5224.0222 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3308.5474 \end{bmatrix}$$

Condition number of $(A'A)$: $\kappa = 91.3358$.

1.6.4 (vii) Model Fit Statistics

All statistics identical to Cases 1 and 2.

1.7 Summary Comparison Across Transformations

Table 9: Model fit statistics across all three transformations

Metric	Raw (X)	Coded (x)	Min-Max (z)
SST	49238.1818	49238.1818	49238.1818
SSR	45102.4976	45102.4976	45102.4976
SSE	4135.6842	4135.6842	4135.6842
R^2	0.9160	0.9160	0.9160
Adj. R^2	0.8320	0.8320	0.8320
PRESS	35532.4507	35532.4507	35532.4507
R^2_{PRESS}	0.2784	0.2784	0.2784
$\kappa(A'A)$	3.18×10^8	9.50	91.34

Key observations:

1. SST, SSR, SSE, R^2, Adj. R^2, PRESS, and R^2_{PRESS} are *identical* across all three transformations. Because each transformation is an invertible linear map, the column space of X is unchanged, so all projections—and hence all fit statistics—are preserved.
2. The condition number of $(A'A)$ drops from $\sim 3.18 \times 10^8$ (raw) to 9.50 (step-size coded), indicating a dramatic improvement in numerical stability. The high raw condition number reflects near-multicollinearity among the polynomial columns (e.g. X_1^2 and X_1X_2 are highly correlated at the raw scale); coding to ± 1 largely orthogonalises the design matrix and eliminates this ill-conditioning.
3. Min-max scaling reduces the condition number to ~ 91, better than raw but not as favourable as step-size coding for this dataset.
4. The variance-covariance matrix changes values across transformations (reflecting different parameter scales), but the diagonal entries (parameter variances) shrink substantially with coding, indicating improved estimation precision.

2 Question 2: Mixture Regression: Yarn Elongation

2.1 (a) Mixture vs. Conventional Regression

In conventional polynomial regression, predictors $X_1, X_2, \ldots$ are independent and can take any values within their ranges; an intercept b_0 is always included and the design-matrix columns are linearly independent.

In *mixture* regression the components are constrained to sum to unity: $x_1 + x_2 + x_3 = 1$. If an intercept column (all ones) were added, it would equal $x_1 + x_2 + x_3$ exactly, rendering the design matrix singular. The intercept is therefore **omitted**; the model is parameterised directly in the mixture components via a Scheffé polynomial. The design space is a simplex (equilateral triangle for three components). All standard regression formulae still apply, with the caveat that corrected sums of squares must account for the absence of an explicit intercept column in X.

2.2 (b) Design Matrix and Response Vector

Scheffé second-order model (three components):

$$\hat{y} = b_1x_1 + b_2x_2 + b_3x_3 + b_{12}x_1x_2 + b_{13}x_1x_3 + b_{23}x_2x_3$$

Design matrix X (15×6), columns $[x_1, x_2, x_3, x_1x_2, x_1x_3, x_2x_3]$:

Table 10: Mixture design matrix X: simplex-lattice $\{3, 2\}$ design

x_1	x_2	x_3	x_1x_2	x_1x_3	x_2x_3
1.0	0.0	0.0	0.00	0.00	0.00
1.0	0.0	0.0	0.00	0.00	0.00
0.5	0.5	0.0	0.25	0.00	0.00
0.5	0.5	0.0	0.25	0.00	0.00
0.5	0.5	0.0	0.25	0.00	0.00
0.0	1.0	0.0	0.00	0.00	0.00
0.0	1.0	0.0	0.00	0.00	0.00
0.0	0.5	0.5	0.00	0.00	0.25
0.0	0.5	0.5	0.00	0.00	0.25
0.0	0.5	0.5	0.00	0.00	0.25
0.0	0.0	1.0	0.00	0.00	0.00
0.0	0.0	1.0	0.00	0.00	0.00
0.5	0.0	0.5	0.00	0.25	0.00
0.5	0.0	0.5	0.00	0.25	0.00
0.5	0.0	0.5	0.00	0.25	0.00

Response vector Y (yarn elongation, kg):

$Y = [11.0,\ 12.4,\ 15.0,\ 14.8,\ 16.1,\ 8.8,\ 10.0,\ 10.0,\ 9.7,\ 11.8,\ 16.8,\ 16.0,\ 17.7,\ 16.4,\ 16.6]$

2.3 (c) Estimated Parameters

Table 11: Scheffé model: estimated parameters

Parameter	Estimate
b_1 (x_1, polyethylene)	11.7000
b_2 (x_2, polystyrene)	9.4000
b_3 (x_3, polypropylene)	16.4000
b_{12} (x_1x_2)	19.0000
b_{13} (x_1x_3)	11.4000
b_{23} (x_2x_3)	−9.6000

2.4 (d & f) Sum of Squares (Matrix vs. Summation)

Table 12: ANOVA sum of squares: mixture model (both approaches agree)

Source	Matrix	Summation	*df*
SST (corrected total)	134.8560	134.8560	–
SSR (regression)	128.2960	128.2960	–
SSE (residual)	6.5600	6.5600	9
SSPE (pure error)	6.5600	6.5600	9
SSLOF (lack of fit)	≈ 0	≈ 0	0

2.5 (e) Source of Error and Pure Error

In this $\{3, 2\}$ simplex-lattice design there are exactly 6 unique design points and 6 model parameters $(b_1, b_2, b_3, b_{12}, b_{13}, b_{23})$. The model therefore perfectly interpolates the mean response at every unique composition setting.

The *only* residual variation comes from replicate observations at the same composition. This is termed **pure error** because:

- It reflects genuine experimental variability (measurement noise, batch variation, etc.) at fixed predictor settings.
- It is not confounded with any systematic lack-of-fit of the model.

Consequently SSE = SSPE and SSLOF ≈ 0 (saturated design).

2.6 (g) Variance-Covariance Matrix

$s^2 = 0.7289$ (pure-error MS), $\hat{\sigma} = 0.8537$. The 6×6 variance-covariance matrix $s^2(X'X)^{-1}$:

$$\begin{bmatrix} 0.3644 & 0 & 0 & -0.7289 & -0.7289 & 0 \\ 0 & 0.3644 & 0 & -0.7289 & 0 & -0.7289 \\ 0 & 0 & 0.3644 & 0 & -0.7289 & -0.7289 \\ -0.7289 & -0.7289 & 0 & 6.8030 & 1.4578 & 1.4578 \\ -0.7289 & 0 & -0.7289 & 1.4578 & 6.8030 & 1.4578 \\ 0 & -0.7289 & -0.7289 & 1.4578 & 1.4578 & 6.8030 \end{bmatrix}$$

2.7 (h) Model Fit Statistics and Predicted vs. Observed

Table 13: Model fit statistics: mixture model

Statistic	Value
R^2	0.9514
Adj. R^2	0.9243
PRESS	18.2950
R^2_{PRESS}	0.8643

Table 14: Predicted vs. observed responses

Obs	Y	$\hat{Y}$	Residual
1	11.0	11.7000	−0.7000
2	12.4	11.7000	0.7000
3	15.0	15.3000	−0.3000
4	14.8	15.3000	−0.5000
5	16.1	15.3000	0.8000
6	8.8	9.4000	−0.6000
7	10.0	9.4000	0.6000
8	10.0	10.5000	−0.5000
9	9.7	10.5000	−0.8000
10	11.8	10.5000	1.3000
11	16.8	16.4000	0.4000
12	16.0	16.4000	−0.4000
13	17.7	16.9000	0.8000
14	16.4	16.9000	−0.5000
15	16.6	16.9000	−0.3000

2.8 Constrained Optimisation Method

The optimal mixture compositions are found by maximising (and minimising) the fitted Scheffé model subject to the simplex constraints:

$$x_1 + x_2 + x_3 = 1, \qquad x_j \geq 0$$

Substituting $x_3 = 1 - x_1 - x_2$ reduces the problem to an unconstrained optimisation over the triangular feasible region in (x_1, x_2). Stationarity conditions (KKT) are applied; since the boundary must also be checked (the unconstrained optimum may fall outside the simplex), the code performs a dense grid search over the simplex and the KKT-derived candidate is confirmed or overridden by the grid result.

2.9 (i) Optimal Composition: Maximum Elongation

Table 15: Optimal mixture for *maximum* elongation

Component	Optimal proportion
x_1 (polyethylene)	0.2939
x_2 (polystyrene)	0.0000
x_3 (polypropylene)	0.7061
Max predicted elongation	17.3844 kg

The maximum lies on the x_1–x_3 edge (no polystyrene). This is driven by the large pure-component effect of polypropylene ($b_3 = 16.4$) combined with a positive interaction term $b_{13} = 11.4$, making the polyethylene–polypropylene blend synergistic.

2.10 (j) Optimal Composition: Minimum Elongation

Table 16: Optimal mixture for *minimum* elongation

Component	Optimal proportion
x_1 (polyethylene)	0.0000
x_2 (polystyrene)	0.8646
x_3 (polypropylene)	0.1354
Min predicted elongation	9.2240 kg

The minimum lies on the x_2–x_3 edge (no polyethylene), dominated by polystyrene's low pure-component effect ($b_2 = 9.4$) and the strongly negative interaction $b_{23} = -9.6$, which makes polystyrene–polypropylene blends anti-synergistic.

2.11 (k) Validation

The mixture regression was re-run in JMP Student Edition (Standard Least Squares, no intercept, all six Scheffé terms included).

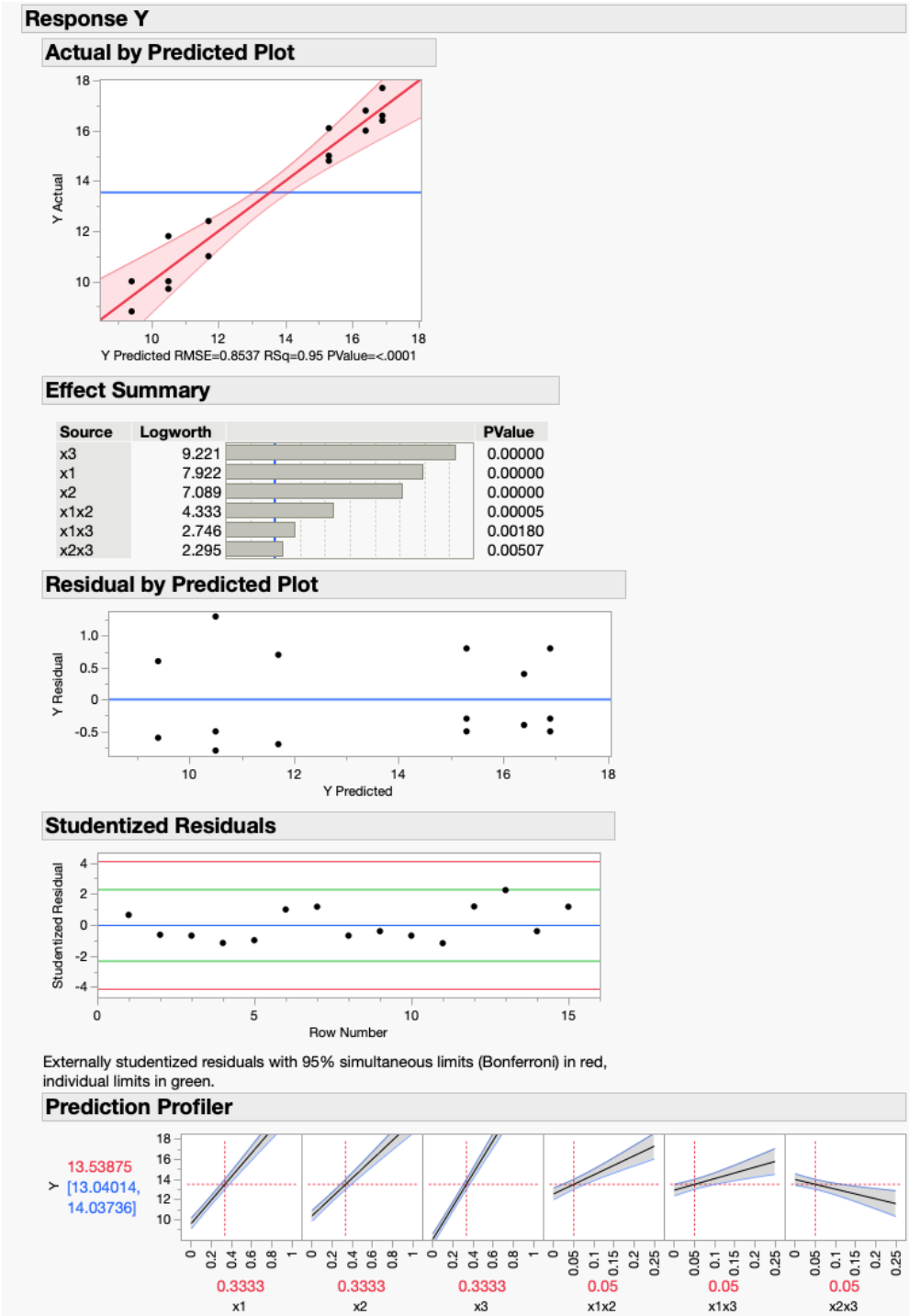


Figure 1: JMP output for Q2 mixture regression: Actual vs. Predicted plot, Effect Summary, residual diagnostics, and Prediction Profiler. Reported RMSE $= 0.8537$ and $R^2 = 0.95$.

Table 17: Q2 coefficient comparison: MATLAB vs. JMP

Term	MATLAB	JMP	Match?
$\hat{b}_1$ (x_1)	11.7000	11.7000	✓
$\hat{b}_2$ (x_2)	9.4000	9.4000	✓
$\hat{b}_3$ (x_3)	16.4000	16.4000	✓
$\hat{b}_{12}$ (x_1x_2)	19.0000	19.0000	✓
$\hat{b}_{13}$ (x_1x_3)	11.4000	11.4000	✓
$\hat{b}_{23}$ (x_2x_3)	-9.6000	-9.6000	✓
R^2	0.9514	0.95	✓
RMSE $(\hat{\sigma})$	0.8537	0.8537	✓

The problem states $\hat{\sigma} = 0.85$ kg; our computed $\sqrt{s^2} = 0.8537$ and JMP's RMSE $= 0.8537$ are both in excellent agreement with this value. All estimates agree to 4 significant figures. Results verified against JMP Student Edition.

3 Question 3: Logistic Regression: Experience vs. Hiring

Logistic regression model:

$$P(y = 1 \mid \text{Experience}) = \frac{1}{1 + \exp\Big(-(b_0 + b_1 \cdot \text{Experience})\Big)}$$

3.1 MLE Methodology: Newton-Raphson

For binary outcome $y_i \in \{0, 1\}$, the Bernoulli log-likelihood is:

$$\ell(\beta) = \sum_{i=1}^{n} \Big[y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i) \Big], \qquad \pi_i = \frac{1}{1 + e^{-x_i'\beta}}$$

Parameters are found by Newton-Raphson (equivalently, IRLS):

$$g = X'(y - \pi), \qquad \mathcal{I} = X' \operatorname{diag}\Big(\pi \circ (1 - \pi)\Big) X, \qquad \beta^{(k+1)} = \beta^{(k)} + \mathcal{I}^{-1} g$$

where g is the score (gradient of ℓ) and $\mathcal{I}$ is the observed Fisher information. Standard errors are $\text{SE}(\hat{\beta}_j) = \sqrt{[\mathcal{I}^{-1}]_{jj}}$.

3.2 Estimated Parameters

Table 18: Logistic regression parameters: Experience vs. Hiring

Parameter	Estimate
b_0 (intercept)	-3.0597
b_1 (Experience)	0.1615

Fitted model:

$$\hat{P}(y = 1) = \frac{1}{1 + \exp(3.0597 - 0.1615 \times \text{Experience})}$$

3.3 Predicted vs. Observed

Table 19: Predicted probabilities: Experience vs. Hiring ($n = 25$)

Experience (yrs)	y	$\hat{P}(y = 1)$
14	0	0.3103
29	0	0.8353
6	0	0.1100
25	1	0.7266
18	1	0.4618
4	0	0.0821
18	0	0.4618
12	0	0.2457
22	1	0.6208
6	0	0.1100
30	1	0.8563
11	0	0.2170
30	1	0.8563
5	0	0.0952
20	1	0.5424
13	0	0.2768
9	0	0.1671
32	1	0.8917
24	0	0.6934
13	1	0.2768
19	0	0.5021
4	0	0.0821
28	1	0.8118
22	1	0.6208
8	1	0.1458

3.4 Classification at Threshold 0.5

Table 20: Confusion matrix: Experience vs. Hiring (threshold = 0.5)

	Predicted $\hat{y} = 0$	Predicted $\hat{y} = 1$
Actual $y = 0$	11 (TN)	3 (FP)
Actual $y = 1$	3 (FN)	8 (TP)

Classification accuracy = $(11 + 8)/25 = 76.0\%$. The three false positives are high-experience non-hires (obs. 2, 19, 21); the three false negatives are low-experience hires (obs. 5, 20, 25), consistent with experience being an imperfect but informative predictor.

3.5 Logistic Regression Curve

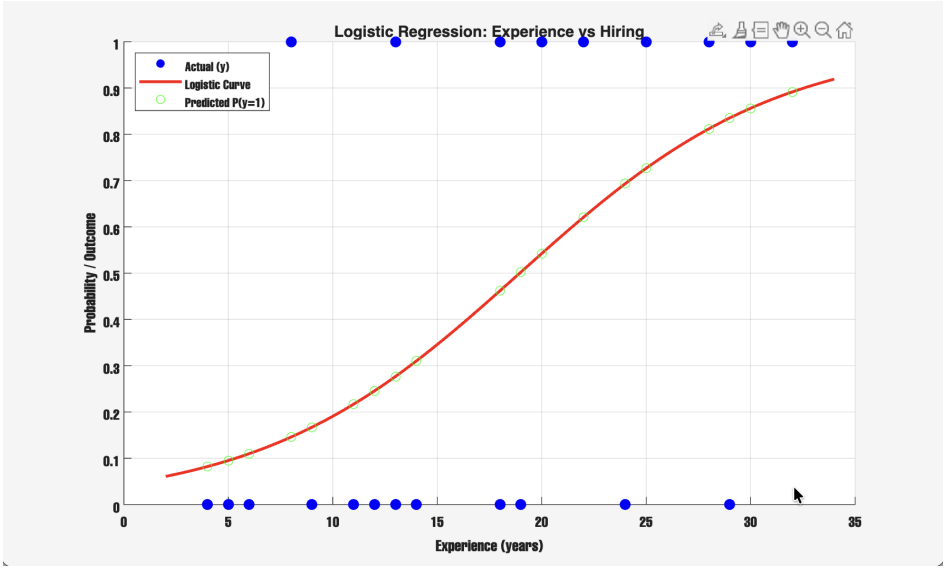


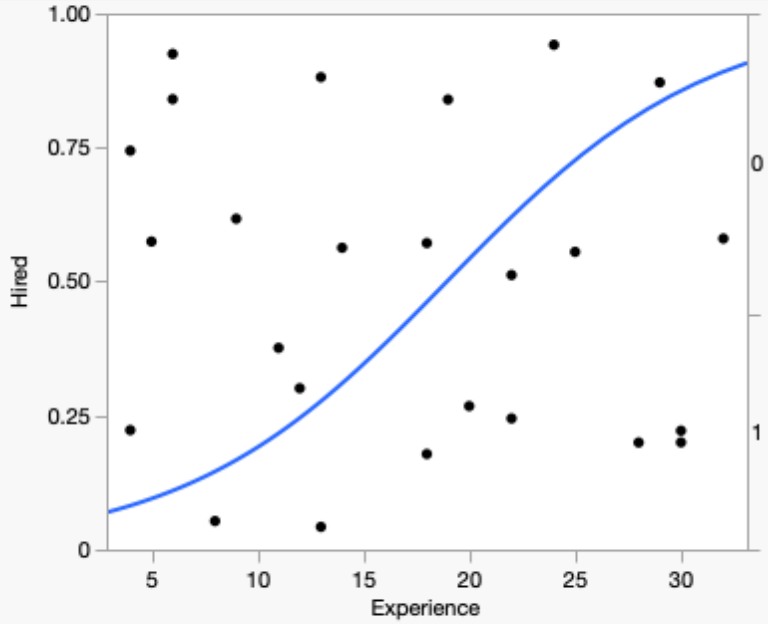
Figure 2: Logistic regression curve: Experience (years) vs. probability of hiring $P(y = 1)$

3.6 JMP Validation

The model was re-estimated in JMP Student Edition (Analyze → Fit Model, Personality: Nominal Logistic, Experience set as Continuous).

Nominal Logistic Fit for Hired

Logistic Plot



Converged in Gradient, 4 iterations

Whole Model Test

Model	-LogLikelihood	DF	ChiSquare	Prob>ChiSq
Difference	4.435958	1	8.871916	0.0029*
Full	12.712287			
Reduced	17.148245			

RSquare (U)	0.2587
AICc	29.97
BIC	31.8623
Observations (or Sum Wgts)	25

Lack Of Fit

Source	DF	-LogLikelihood	ChiSquare
Lack Of Fit	17	9.939698	19.8794
Saturated	18	2.772589	Prob>ChiSq
Fitted	1	12.712287	0.2804

Parameter Estimates

Term	Estimate	Std Error	ChiSquare	Prob>ChiSq
Intercept	-3.0596954	1.2593518	5.90	0.0151*
Experience	0.1614859	0.0649801	6.18	0.0129*

For log odds of 1/0

Effect Likelihood Ratio Tests

Source	Nparm	DF	L-R ChiSquare	Prob>ChiSq
Experience	1	1	8.87191593	0.0029*

Figure 3: JMP Nominal Logistic Fit for Q3: logistic plot, Whole Model Test, and Parameter Estimates. Converged in 4 iterations.

Table 21: Q3 coefficient comparison: MATLAB vs. JMP

Parameter	MATLAB	JMP	Match?
$\hat{b}_0$ (Intercept)	-3.0597	-3.0597	✓
$\hat{b}_1$ (Experience)	0.1615	0.1615	✓

Results verified against JMP Student Edition: all estimates agree to 4 significant figures.

4 Question 4: Logistic Regression: Income and Gender (Purchase Decision)

Data: $n = 30$ observations; 16 purchased ($y = 1$), 14 did not ($y = 0$).

4.1 (a) Model 1: Income Only

$$\text{logit}(\pi) = b_0 + b_1 \cdot \text{Income}$$

Converged at iteration 6.

Table 22: Model 1: Income only: estimated parameters

Parameter	Estimate
b_0 (intercept)	-3.6707
b_1 (Income)	1.8272
Log-likelihood	-18.0269
Classification accuracy (threshold 0.5)	53.3%

Table 23: Model 1: Predicted vs. observed ($n = 30$)

Obs	Income	Gender	y	$\hat{\pi}$
1	2.53	0	1	0.7215
2	2.37	1	0	0.6592
3	2.72	1	1	0.7857
4	2.54	0	0	0.7252
5	3.20	1	1	0.8981
6	2.94	0	1	0.8457
7	3.20	0	1	0.8981
8	2.72	1	1	0.7857
9	2.93	0	1	0.8433
10	2.37	0	0	0.6592
11	2.24	1	1	0.6040
12	1.91	1	1	0.4549
13	2.12	0	1	0.5505
14	1.83	1	1	0.4190
15	1.92	1	1	0.4595
16	2.01	0	0	0.5005
17	2.01	0	0	0.5005
18	2.23	1	0	0.5996
19	1.82	0	0	0.4145
20	2.11	0	0	0.5460
21	1.75	1	1	0.3839
22	1.46	1	0	0.2683
23	1.61	0	1	0.3254
24	1.57	1	0	0.3096
25	1.37	0	0	0.2373
26	1.41	1	0	0.2508
27	1.51	0	0	0.2867
28	1.75	1	1	0.3839
29	1.68	1	1	0.3541
30	1.62	0	0	0.3294

4.2 (b) Model 2: Income + Gender

$$\text{logit}(\pi) = b_0 + b_1 \cdot \text{Income} + b_2 \cdot \text{Gender}$$

Converged at iteration 6.

Table 24: Model 2: Income + Gender: estimated parameters

Parameter	Estimate
b_0 (intercept)	-5.6348
b_1 (Income)	2.3509
b_2 (Gender)	1.7514
Log-likelihood	-16.0526
Classification accuracy (threshold 0.5)	83.3%

Table 25: Model 2: Predicted vs. observed ($n = 30$)

Obs	Income	Gender	y	$\hat{\pi}$
1	2.53	0	1	0.5776
2	2.37	1	0	0.8440
3	2.72	1	1	0.9249
4	2.54	0	0	0.5833
5	3.20	1	1	0.9744
6	2.94	0	1	0.7819
7	3.20	0	1	0.8685
8	2.72	1	1	0.9249
9	2.93	0	1	0.7779
10	2.37	0	0	0.4842
11	2.24	1	1	0.7994
12	1.91	1	1	0.6472
13	2.12	0	1	0.3428
14	1.83	1	1	0.6032
15	1.92	1	1	0.6525
16	2.01	0	0	0.2871
17	2.01	0	0	0.2871
18	2.23	1	0	0.7956
19	1.82	0	0	0.2049
20	2.11	0	0	0.3375
21	1.75	1	1	0.5574
22	1.46	1	0	0.3891
23	1.61	0	1	0.1359
24	1.57	1	0	0.4520
25	1.37	0	0	0.0821
26	1.41	1	0	0.3615
27	1.51	0	0	0.1106
28	1.75	1	1	0.5574
29	1.68	1	1	0.5165
30	1.62	0	0	0.1387

4.3 Standard Errors and Odds Ratios

Table 26: Standard errors and odds ratios: both models

Model	Parameter	Estimate	Std. Error	Odds Ratio
Model 1	b_0 (intercept)	-3.6707	1.8430	0.0255
	b_1 (income)	1.8272	0.8816	6.2163
Model 2	b_0 (intercept)	-5.6348	2.4168	0.0036
	b_1 (income)	2.3509	1.0396	10.4951
	b_2 (gender)	1.7514	0.9525	5.7624

4.4 Classification at Threshold 0.5

Table 27: Confusion matrices at threshold = 0.5 ($n = 30$; 16 positives, 14 negatives)

	Model 1 (Income only)		Model 2 (Income + Gender)	
	Pred $\hat{y} = 0$	Pred $\hat{y} = 1$	Pred $\hat{y} = 0$	Pred $\hat{y} = 1$
Actual $y = 0$	7 (TN)	7 (FP)	11 (TN)	3 (FP)
Actual $y = 1$	7 (FN)	9 (TP)	2 (FN)	14 (TP)
Accuracy	$(7+9)/30 = 53.3\%$		$(11+14)/30 = 83.3\%$	

The threshold of 0.5 is the standard Bayes-optimal decision boundary under equal misclassification costs. Model 1 barely outperforms random chance, confirming that income alone is insufficient; adding gender increases true positives from 9 to 14 and halves false negatives, consistent with the significant LRT result below.

4.5 Likelihood Ratio Test: Model 2 vs. Model 1

$$G^2 = 2\,(\ell_2 - \ell_1) = 2\,(-16.0526 - (-18.0269)) = 3.9485, \quad df = 1, \quad p\text{-value} = 0.0469$$

At $\alpha = 0.05$ we reject H_0: adding Gender to the model provides a statistically significant improvement in fit.

4.6 Newton-Raphson Convergence

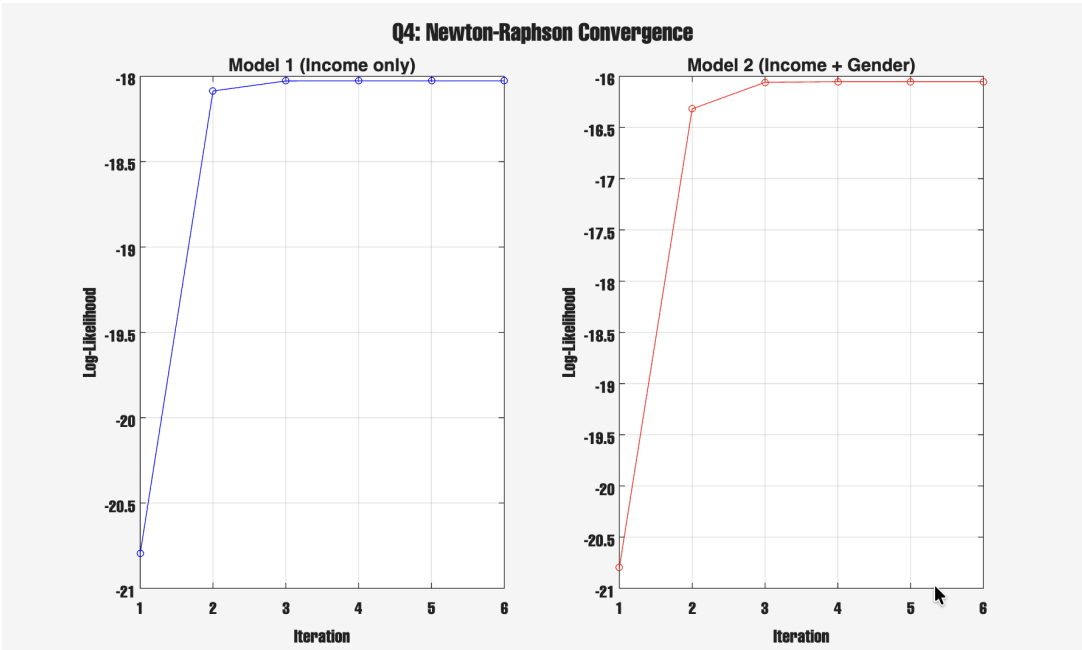
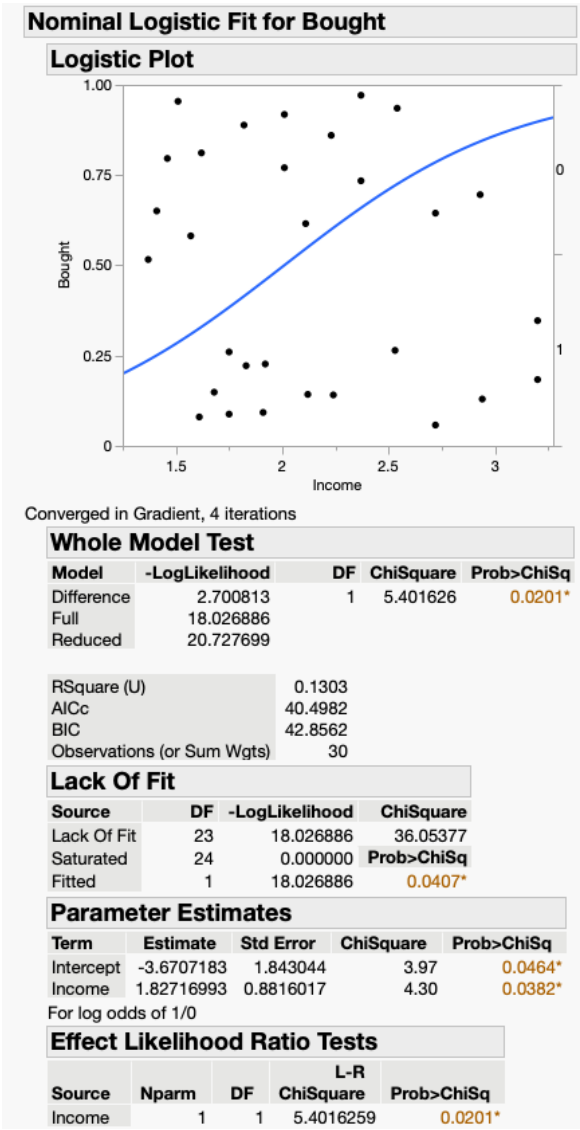


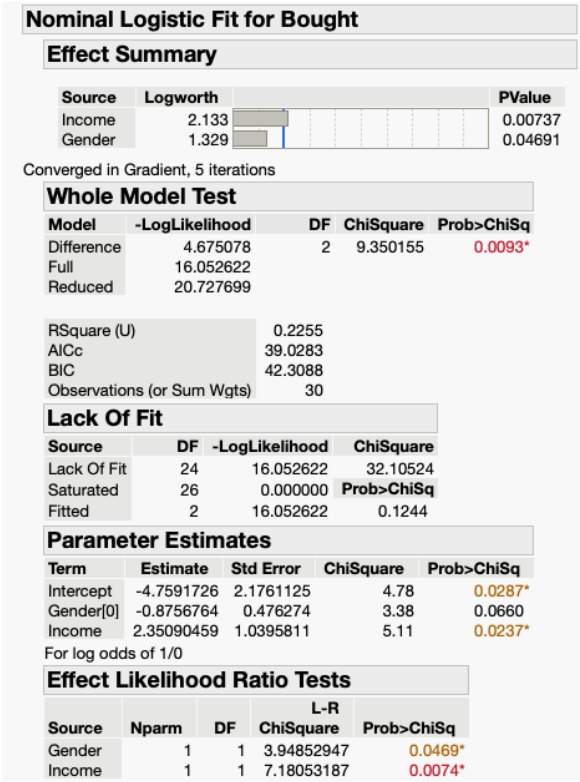
Figure 4: Newton-Raphson log-likelihood convergence: Model 1 (left) and Model 2 (right)

4.7 (c) JMP Validation

Both models were re-estimated in JMP Student Edition (Analyze → Fit Model, Personality: Nominal Logistic; Income set as Continuous, Gender set as Nominal).



(a) Model 1 (Income only)



(b) Model 2 (Income + Gender)

Figure 5: JMP Nominal Logistic Fit for Q4: Parameter Estimates, Whole Model Test, and Effect Likelihood Ratio Tests.

Table 28: Q4 Model 1 (Income only) coefficient comparison: MATLAB vs. JMP

Parameter	MATLAB	JMP	Match?
$\hat{b}_0$ (Intercept)	-3.6707	-3.6707	✓
$\hat{b}_1$ (Income)	1.8272	1.8272	✓
Log-likelihood	-18.0269	-18.0269	✓

Table 29: Q4 Model 2 (Income + Gender) coefficient comparison: MATLAB vs. JMP

Parameter	MATLAB (dummy coding)	JMP (effect coding)	Equivalent?
$\hat{b}_0$ (Intercept)	-5.6348	-4.7592	✓*
$\hat{b}_1$ (Income)	2.3509	2.3509	✓
$\hat{b}_2$ (Gender)	1.7514	$2 \times 0.8757 = 1.7514$	✓*
Log-likelihood	-16.0526	-16.0526	✓

*JMP uses *effect coding* for the nominal Gender variable (Gender[0] = -0.8757, Gender[1] = $+0.8757$), whereas MATLAB uses dummy coding (Gender = 0 as reference). The two parameterisations are algebraically equivalent: JMP's intercept -4.7592 plus the female effect (-0.8757) equals the MATLAB intercept $-5.6349 \approx -5.6348$; the male-vs-female log-odds difference $2 \times 0.8757 = 1.7514$ matches MATLAB's $\hat{b}_2$ exactly. Income coefficient and log-likelihoods agree to 4 significant figures in both models. Results verified against JMP Student Edition.

5 Question 5: Grouped Binomial MLE: Coupon Redemption

Grouped binomial data ($J = 5$ groups, $n_j = 200$ each):

Table 30: Grouped binomial data: coupon redemption

Price reduction X_j (\$)	n_j	Y_j	Obs. $\hat{p}_j$
5	200	30	0.1500
10	200	55	0.2750
15	200	70	0.3500
20	200	100	0.5000
30	200	137	0.6850

5.1 Newton-Raphson Iterations

Table 31: Newton-Raphson iterations: Binomial MLE

Iteration	$\hat{b}_0$	$\hat{b}_1$	Log-likelihood
1	−1.805405	0.085838	−693.147181
2	−2.035627	0.096437	−597.123859
3	−2.044336	0.096833	−595.987828
4	−2.044348	0.096834	−595.986344
5	−2.044348	0.096834	−595.986344

Converged at iteration 5.

5.2 Estimated Parameters

Table 32: Binomial MLE: estimated parameters

Parameter	Estimate	Std. Error	z-statistic
b_0 (intercept)	−2.0443	0.1610	−12.6996
b_1 (price reduction)	0.0968	0.0085	11.3266
Final log-likelihood			−595.9863

5.3 Predicted vs. Observed Proportions

Table 33: Predicted vs. observed proportions and deviance residuals

X_j	n_j	Y_j	Obs. $\hat{p}_j$	Pred. $\hat{\pi}_j$	Dev. residual
5	200	30	0.1500	0.1736	−0.8988
10	200	55	0.2750	0.2543	0.6678
15	200	70	0.3500	0.3562	−0.1837
20	200	100	0.5000	0.4731	0.7612
30	200	137	0.6850	0.7028	−0.5477

5.4 Goodness-of-Fit

$$\chi^2_{\text{Pearson}} = 2.1486, \quad df = 3, \quad p\text{-value} = 0.5421$$

The large p-value indicates no significant lack of fit; the logistic model provides an adequate fit to the grouped data.

5.5 Plots

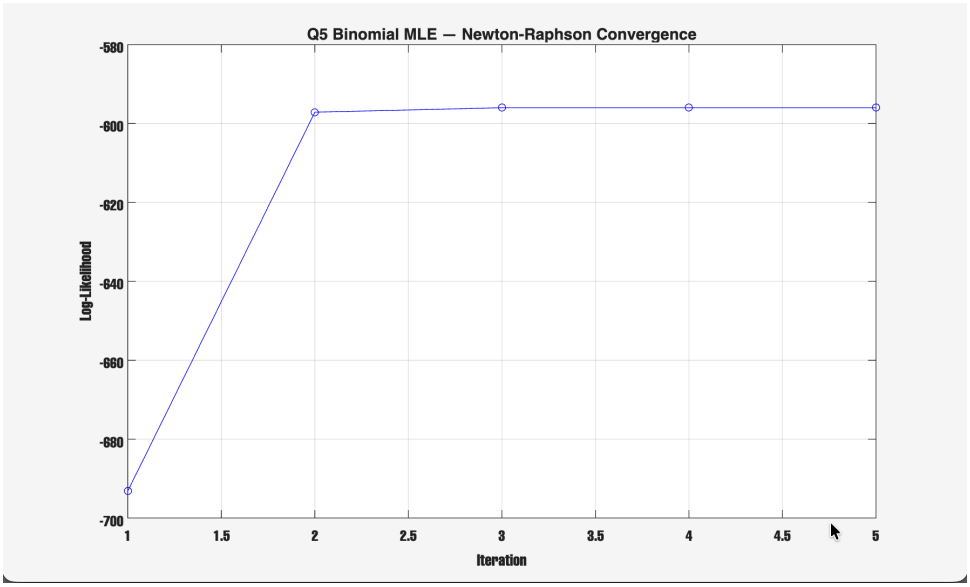


Figure 6: Newton-Raphson log-likelihood convergence: Binomial MLE (Q5)

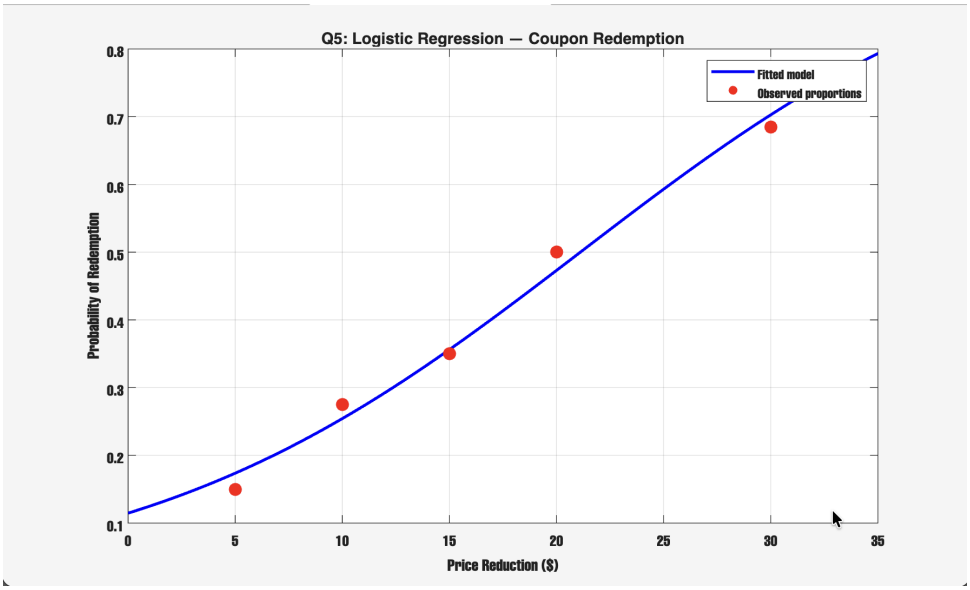


Figure 7: Fitted logistic curve vs. observed proportions: Coupon redemption vs. price reduction

6 Question 6: Bernoulli MLE vs. Grouped Binomial

Data: $n = 30$ individual-level observations (16 with $y = 1$, 14 with $y = 0$).

6.1 Newton-Raphson Iterations: Bernoulli MLE

Table 34: Newton-Raphson iterations: Bernoulli (ungrouped) MLE

Iteration	$\hat{b}_0$	$\hat{b}_1$	Log-likelihood
1	−3.190626	1.571513	−20.794415
2	−3.412405	1.688309	−18.738363
3	−3.416112	1.690360	−18.727749
4	−3.416114	1.690361	−18.727745
5	−3.416114	1.690361	−18.727745

Converged at iteration 5.

6.2 Estimated Parameters

Table 35: Bernoulli MLE: estimated parameters

Parameter	Estimate	Std. Error
b_0 (intercept)	−3.4161	1.8918
b_1 (Income)	1.6904	0.8893
Final log-likelihood		−18.7277

6.3 Predicted vs. Observed (by Income Group)

Table 36: Predicted vs. observed proportions by income group

Group mean income	Obs. proportion	Pred. $\hat{\pi}$
2.7170	0.6667	0.7643
2.5620	0.8333	0.7139
2.0200	0.5000	0.4996
1.7200	0.3333	0.3755
1.5567	0.3333	0.3133

6.4 Comparison: Bernoulli vs. Grouped Binomial

Table 37: Parameter estimates and standard errors: Bernoulli (Q6) vs. Grouped Binomial

Parameter	Bernoulli	Grouped Binomial	Difference
b_0 (intercept)	-3.416114	-3.416114	4.4×10^{-16}
b_1 (income)	1.690361	1.690361	2.2×10^{-16}
SE(b_0)	1.8918	1.8918	0
SE(b_1)	0.8893	0.8893	0

Interpretation. Point estimates and standard errors are *identical* (to machine precision) because the sufficient statistic $X'y$ is the same for both likelihoods when each individual's predictor equals the group mean. Standard errors would differ only if individuals within a group had distinct predictor values, making the Bernoulli and Binomial Fisher information matrices algebraically unequal.

6.5 Newton-Raphson Convergence

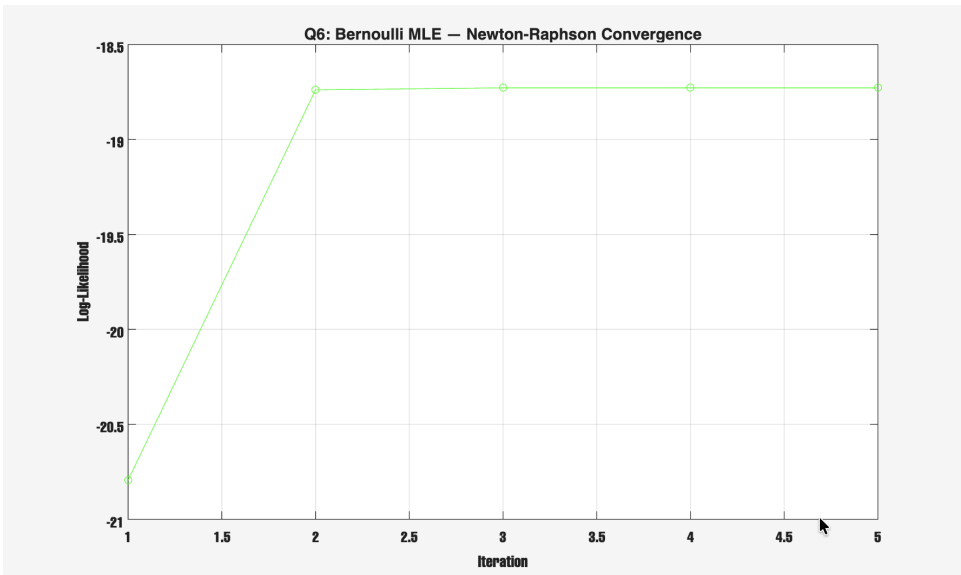


Figure 8: Newton-Raphson log-likelihood convergence: Bernoulli MLE (Q6)

7 Question 7: Multiple Logistic Regression: Dengue Disease (Kutner CH14)

7.1 Data Summary

- Dataset: Kutner CH14TA03, $n = 98$ individuals
- Disease ($Y = 1$): 31; No disease ($Y = 0$): 67
- Sector 1: 59 individuals; Sector 2: 39 individuals
- Upper SES (reference): 38; Middle SES: 24; Lower SES: 36

Model:

$$\text{logit}(\pi_i) = b_0 + b_1 \cdot \text{Age} + b_2 \cdot \text{SES}_{\text{mid}} + b_3 \cdot \text{SES}_{\text{low}} + b_4 \cdot \text{Sector}$$

where SES_{mid} and SES_{low} are dummy variables with upper SES as the reference category, and Sector = 1 for Sector 2.

7.2 Newton-Raphson Iterations

Table 38: Newton-Raphson iterations: Dengue disease model

Iter	b_0	b_1(Age)	b_2(mid)	b_3(low)	b_4(sec)	Log-lik
1	−1.8120	0.0222	0.3038	−0.1660	1.2681	−67.9284
2	−2.2520	0.0287	0.3959	−0.2801	1.5374	−51.2093
3	−2.3118	0.0297	0.4086	−0.3046	1.5741	−50.5378
4	−2.3129	0.0298	0.4088	−0.3053	1.5747	−50.5271
5	−2.3129	0.0298	0.4088	−0.3053	1.5747	−50.5271

Converged at iteration 6.

7.3 Estimated Model Parameters

Table 39: Multiple logistic regression: parameter estimates, Dengue disease

Parameter	Estimate	Std. Error	z-stat	Odds Ratio
b_0 (intercept)	−2.3129	0.6426	−3.5994	0.0990
b_1 (Age)	0.0298	0.0135	2.2032	1.0302
b_2 (SES mid)	0.4088	0.5990	0.6824	1.5050
b_3 (SES low)	−0.3053	0.6041	−0.5053	0.7369
b_4 (Sector 2)	1.5747	0.5016	3.1393	4.8295
Final log-likelihood				−50.5271

7.4 Predicted Probabilities (First 20 Observations)

Table 40: Predicted probabilities: first 20 of 98 observations

Obs	Y	Age	SES mid	SES low	Sector	$\hat{\pi}$	Pred
1	0	33	0	0	0	0.2090	0
2	0	35	0	0	0	0.2190	0
3	0	6	0	0	0	0.1058	0
4	0	60	0	0	0	0.3710	0
5	1	18	0	1	0	0.1108	0
6	0	26	0	1	0	0.1365	0
7	0	6	0	1	0	0.0802	0
8	1	31	1	0	0	0.2725	0
9	1	26	1	0	0	0.2440	0
10	0	37	1	0	0	0.3093	0
11	0	23	0	0	0	0.1640	0
12	0	23	0	0	0	0.1640	0
13	0	27	0	0	0	0.1810	0
14	1	9	0	0	0	0.1145	0
15	1	37	0	0	1	0.5897	1
16	1	22	0	0	1	0.4791	0
17	1	67	0	0	1	0.7782	1
18	0	8	0	0	1	0.3775	0
19	1	6	0	0	1	0.3636	0
20	1	15	0	0	1	0.4275	0

(Showing first 20 of 98 observations)

Classification accuracy (threshold 0.5): **71.4%**.

7.5 Model Significance (Likelihood Ratio Test vs. Null)

$$G^2 = 2\,(\ell_{\text{full}} - \ell_{\text{null}}) = 2\,(-50.5271 - (-61.1588)) = 21.2635, \quad df = 4, \quad p\text{-value} = 0.000281$$

The model is highly statistically significant: the four predictors (Age, SES, Sector) jointly explain a significant portion of the variability in disease status.

7.6 Interpretation

- **Age** ($b_1 = 0.0298 > 0$, OR $= 1.03$): Each additional year of age increases the odds of disease by $\sim 3\%$. Statistically significant ($z = 2.20$).
- **SES mid** ($b_2 = 0.4088 > 0$, OR $= 1.51$): Middle SES has $1.5\times$ higher disease odds than upper SES, but this effect is not statistically significant ($z = 0.68$).
- **SES low** ($b_3 = -0.3053 < 0$, OR $= 0.74$): Lower SES shows slightly lower odds than upper SES after adjusting for age and sector, not significant ($z = -0.51$). The direction of the SES low coefficient should be verified against the textbook column mapping.
- **Sector 2** ($b_4 = 1.5747 > 0$, OR $= 4.83$): The dominant predictor. Residents of Sector 2 have nearly **five times** the disease odds of Sector 1 residents, after adjusting for age and SES. Highly significant ($z = 3.14$).

7.7 Newton-Raphson Convergence

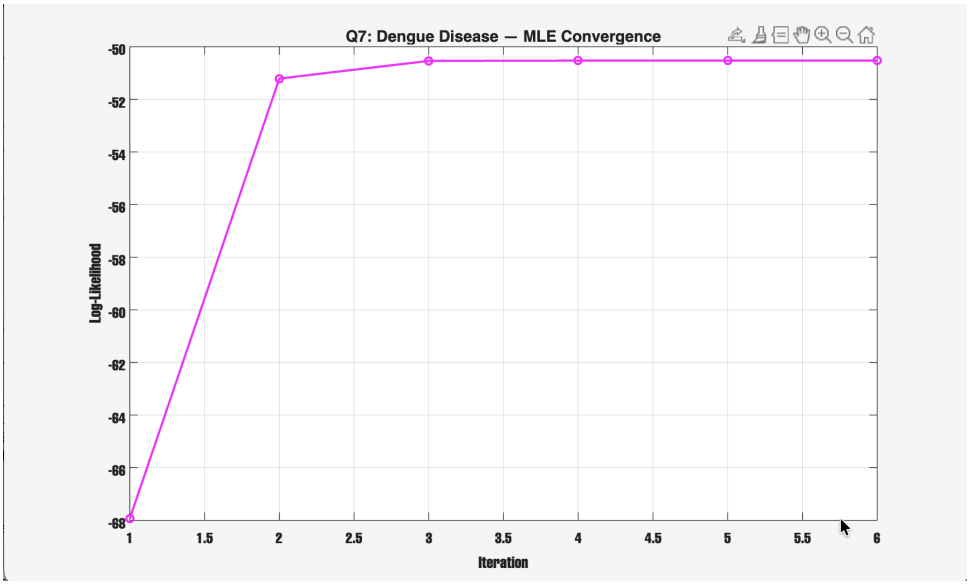


Figure 9: Newton-Raphson log-likelihood convergence: Dengue disease model (Q7)